Neha Mungale

Email — Mobile — LinkedIn — GitHub — Website

EDUCATION

University College London, *London, UK* Sep 2025 – Sep 2026
M.Sc. in Artificial Intelligence for Biomedicine and Healthcare

Indian Institute of Technology Delhi, *Delhi, India* Aug 2021 – May 2025
B.Tech in Biochemical Engineering & Biotechnology GPA: **8.1**/10 – Dept. Rank: **7/70+**

PROFESSIONAL EXPERIENCE

GSK Jul 2026 – Present
Decision Sciences and AI Intern, Enterprise AI Team London, UK

- One of 3 forward deployed engineers to build a **payroll reconciliation platform** across 30+ countries and multiple vendors, to **prevent financial leakage** and fraud, autogenerate **SOX compliance evidence**, and surface where global payroll processes actually break; owned the frontend end-to-end and contributed across the full stack, demoing weekly to stakeholders.
- Built reusable **AI tooling** that multiplied team throughput: generating brand compliant screens (**1 to 4 screens** in the same time); auditing persona access gaps against the live codebase to cut back and forth in UAT; E2E testing in CI to verify real user workflows.
- Contributed to an agentic AI harness for autonomous software development workflows, with legal, compliance and product oversight.

Indian Institute of Science (IISc) – collab. Indian Council of Medical Research Jun 2025 – Aug 2025
Analytics Intern, Centre of Data for Public Good (CDPG) Bangalore, India

- Built **clinical data ingestion and validation pipelines** for ADARV, a national public health analytics platform, ensuring FAIR-compliant data collection.
- Contributed to an LLM **Q&A system** (DeepSeek + FAISS) over an **FHIR** knowledge graph, and an NLP entity mapping pipeline (spaCy, RapidFuzz) for automated **SNOMED CT** standardisation.

CentraLogic May 2024 – Jul 2024
AI/ML Intern Pune, India

- Took an LLM proof-of-concept to a **production user facing application**: engineered an end-to-end FastAPI microservice (Pydantic validation, API versioning) with a Streamlit frontend on locally hosted, quantised LLMs; built **LangGraph RAG pipelines** (Meta Llama 3, Hugging Face) for multi-source document reasoning.

University of Oxford – Roth Lab Jun 2023 – Jul 2023
Research Intern Oxford, UK

- Designed wet lab experiments and quantified fungal root colonisation by bright field microscopy in wildtype vs. an **amiRNA-TET8** knockdown line (9 timepoints, 10 replicates); analysed time series data in R (Shapiro-Wilk, Dunn's).

RESEARCH EXPERIENCE

Self Driving Labs Apr 2026 – Sep 2026
MSc Thesis (UCL), Supervisor: Dr. David Shorthouse GitHub

- Built a **physical closed loop pilot** (OT-2 liquid handler, xArm5 arm, SpectraMax, ROS2): **ArUco guided** registration and **motion planning**, validated in a **Gazebo digital twin** of the lab setup, for reliable robotic arm navigation between instruments.
- First systematic test of how much agency an LLM should get in **Bayesian optimisation** for **noisy, short horizon, multi-objective** formulation, where a failed 40 experiment campaign cannot be re-run: warm start priors improves reliability, continuous LLM control can degrade it (5 datasets).
- Built **GP surrogate diagnostics** that catch silent model failure at low sample sizes, and a pre-run checklist explaining why **LLM evolved acquisition functions** work on synthetic benchmarks but fail to generalise on noisy real domains.

RLVR & LLM Fine-Tuning: Reasoning with Composite Reward Functions Feb 2026 – Mar 2026
Graduate Coursework (MSc, UCL) London, UK

- Implemented **SFT with LoRA/QLoRA** on Gemma-2B and a composite **GRPO/PPO** reward function.
- Analysed failure modes across SFT-initialised vs. base-model RLVR training strategies.

Accountable Deep Learning for Diabetic Retinopathy Triage Nov 2025 – Dec 2025
Graduate Project (MSc, UCL) GitHub

- Achieved **QWK 0.916** on 3,600+ fundus images (PyTorch) using Deep Ordinal Regression; integrated uncertainty quantification and SmoothGrad-IG attribution maps for auditable predictions.

Hypoxia Based Risk Stratification in Meningioma Jul 2024 – May 2025
B.Tech Thesis (IIT Delhi), Supervisor: Prof. Ritu Kulshreshtha (HOD)

- Built an end-to-end R pipeline to derive a **prognostic biomarker** for hypoxia driven risk stratification across 5 transcriptomic cohorts (n = 1,032; DESeq2, GSVA, fgsea, CIBERSORTx, WGCNA); AUC = 84.4%, C-index = 0.837; **LASSO** recurrence model.

SKILLS

- **Programming:** Python, R, SQL, MATLAB, Git/GitHub
- **Robotics:** ROS2, Opentrons, xArm5, Gazebo, Orbbec Femto Bolt RGB-D camera
- **Software & Systems:** FastAPI, Pydantic, Docker, CI/CD, knowledge graphs (Neo4j, Apache AGE, Cypher), LangChain, LangGraph, RAG, LLM/agent evals
- **ML & Bioinformatics:** PyTorch, scikit-learn, Hugging Face, DESeq2, GSVA, fgsea, CIBERSORTx, WGCNA

SCHOLASTIC ACHIEVEMENTS

- **Roche Mentorship Programme 2026:** Selected to benchmark readiness against industry hiring standards in the pharmaceutical sector.
- **DST INSPIRE Scholar:** Selected as top 1% of science students by India's Dept. of Science & Technology.
- **JEE Advanced 2021:** All-India Rank **3770** ($\sim$160,000 candidates).
- **JEE Mains 2021:** Percentile **99.15** among $\sim$1 million candidates.